ADEMP protocol: 09-informative-dropout-by-design

2026-05-07 17:00 PDT

This document is the working protocol for the exploratory simulation study that underlies `analysis/report/09-informative-dropout-by-design/`. The study is exploratory, and its findings should be confirmed in an independent study. ADEMP structure follows Morris, White and Crowther (2019).

Common infrastructure

Trial designs (four-design factorial).

1. *Open-label* (OL): 20 weeks of open-label active drug.
2. *Open-label plus blinded discontinuation* (OL+BDC): 16 weeks of OL active + 4 weeks of blinded discontinuation.
3. *Traditional crossover* (CO): 10 weeks period 1 + 10 weeks period 2, with within-participant randomisation.
4. *Hybrid N-of-1*: 8 weeks OL + 4 weeks BDC + 4 weeks crossover-1 + 4 weeks crossover-2 (the Hendrickson 2020 primary design).

All four designs at 20 weeks total to match Hendrickson 2020 Figure 2.

Sample sizes. $N \in \{35, 70, 100\}$ per design.

Drug context. Prazosin for PTSD nightmares, calibrated to the Hendrickson 2020 reference parameters.

Carryover. Exponential decay with $t_{1/2} = 3$ days at the baseline cell.

Replicates. $n_{\text{reps}} = 1000$ per cell for power and bias estimands; $n_{\text{reps}} = 5000$ for type I error cells.

Random-number control. Per-replicate seed via cell-descriptor hash; base seed in `01-base-seed.txt`.

Reporting standard. Morris-White-Crowther ADEMP with **MCSE columns alongside every reported performance measure**.

Study 1. Power × design × dropout pattern

A. Reproduce and extend Hendrickson 2020 Figure 4A: power as a function of (design family) × (dropout pattern) × (sample size) × (biomarker effect size).

D. Three-component DGP from the pmsimstats simulation engine with the @hendrickson2020 hazard-based dropout layer. Five dropout patterns:

1. *No dropout* (reference): $\beta_0 = 0$, $\beta_1 = 0$.
2. *Balanced*: $\beta_0 = 0.05$, $\beta_1 = 0.5$ (Hendrickson default).
3. *More-of-flat*: $\beta_0 = 0.05$, $\beta_1 = 0.2$ (treatment non-responders drop more).
4. *More-of-biased*: $\beta_0 = 0.05$, $\beta_1 = 0.8$ (treatment-responders drop more on discontinuation).
5. *High-dropout*: $\beta_0 = 0.15$, $\beta_1 = 0.5$ (overall higher rate, balanced direction).

The hazard at each timepoint is $\Pr(\text{drop at } t) = \beta_0 + \beta_1 \cdot \Delta_{Sx}(t)^2$ where Δ_{Sx} is the change in symptom score since baseline scaled to $[0, 100]$.

E. Primary estimand: power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ in the linear-mixed analysis. Secondary estimand: bias of $\hat{\beta}_{bm:D}$ relative to the known-true effect at each cell.

M. Linear-mixed analysis `Sx ~ bm + t + Dbc + bm:Dbc + (1 | ptID)` with `corCAR1(form = ~t|ptID)`, fitted to the post-dropout retained data. Mixed-effects ML missing-data inference applies under the assumption of MAR; we explicitly note that under our DGP dropout depends on observed symptom trajectory, which gives MNAR semantics relative to the unobserved future trajectory but MAR semantics relative to the observed past.

P. Power, type I error under $\beta_{bm:D} = 0$, bias, empirical Monte Carlo SE of $\hat{\beta}_{bm:D}$ across replicates, model SE, nominal-95% CI coverage, convergence rate, mean dropout fraction. **MCSE attached to every reported estimate**, computed as $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$ for proportions and the standard Monte Carlo SE of the within-replicate mean for continuous measures.

Working prediction. Power loss from dropout exceeds 50% in the most-vulnerable design-by-dropout cells (high dropout in OL+BDC and N-of-1 designs) and remains negligible in the OL design across all dropout patterns. Design rankings on power reverse depending on the dropout pattern: OL+BDC and N-of-1 dominate under no dropout but fall below CO under symptom-improvement-driven dropout.

Study 2. Bias quantification (two-mode)

A. Quantify the bias of the biomarker-treatment interaction estimate under each dropout pattern, in two reference modes: (a) bias relative to the known true value at the cell, (b) bias relative to the gold-standard estimate from the full uncensored dataset.

D. As Study 1, but with bias as the primary estimand and the $c_{bm} \in \{0.3, 0.6\}$ cells the focus.

E. $\Delta\beta = \hat{\beta}^{\text{dropped}}_{bm:D} - \beta^{\text{true}}_{bm:D}$ (mode a) and $\Delta\beta = \hat{\beta}^{\text{dropped}}_{bm:D} - \hat{\beta}^{\text{gold}}_{bm:D}$ (mode b), where $\hat{\beta}^{\text{gold}}_{bm:D}$ is the analysis-model estimate from the full uncensored data.

M. As Study 1.

P. Mean and Monte Carlo SE of $\Delta\beta$ across replicates; the proportion of replicates where $\Delta\beta$ crosses zero (effect-direction reversal under dropout).

Study 3. Randomization-path 'happy accident' decomposition

A. Quantify the @hendrickson2020 'happy accident' selection effect: in OL+BDC and hybrid N-of-1 designs, the multiple randomization paths interact with biased dropout to preferentially preserve the most-informative crossover blocks.

D. Three-component DGP at the $c_{bm} = 0.3$ effect size, with the OL+BDC and hybrid N-of-1 designs only, under the balanced and more-of-biased dropout patterns.

E. Power conditional on randomization path. Within OL+BDC, the path indicator records which 1-week sub-block of the 4-week BDC phase the participant was discontinued; within hybrid N-of-1, the indicator records the on/off pattern across the two crossover blocks.

M. As Study 1, with path-conditional analysis.

P. Path-conditional power; the difference between the weighted average power (weighted by the empirical path distribution under dropout) and the unweighted average power (weighted by the design's nominal path distribution). The difference quantifies the 'happy accident' selection effect.

Study 4. Sensitivity to dropout-pattern misspecification

A. Quantify the inferential cost of analysing the data under an incorrect missing-data assumption (MCAR when truth is MNAR; MAR with the wrong covariate structure).

D. As Studies 1-2, with the analysis-side missing-data treatment varied:

1. Naive complete-case (drop participants on first missing).
2. MAR-conditional MMRM (the Study 1 analysis).
3. Multiple-imputation under the correct MNAR mechanism.

E. Bias and power as in Studies 1 and 2.

M. Three analysis variants per the D specification.

P. Power, bias, MCSE, coverage. The differences between the three analysis variants quantify the cost of mis-handling the missing-data mechanism.

Total cell counts

- Study 1: 4 designs × 5 patterns × $3N$ × $3c_{bm}$ = 180 cells.
- Study 2: subset of Study 1 at $c_{bm} \in \{0.3, 0.6\}$ = 120 cells.
- Study 3: 2 designs × 2 patterns × $3N$ = 12 cells, with path-conditional decomposition adding per-cell complexity.
- Study 4: 4 designs × 5 patterns × 3 analyses × N = 35 = 60 cells.

Deliverables

- Driver scripts at `analysis/scripts/informative-dropout-by-design/`.
- ADEMP results tables per study **with MCSE columns alongside every estimate** (per Morris recommendation, addressing the manuscript-stage gap flagged in the project ADEMP audit).
- Figures: power-by-(design, dropout-pattern) heatmap (Study 1 reproduces Hendrickson Figure 4A and extends); bias plots (Study 2); path-conditional power decomposition (Study 3); analysis-method sensitivity (Study 4).

Protocol change log

`02-deviations-log.md` records changes to the cells, estimands, methods, or performance measures specified in this working protocol.

Source: `~/prj/alz/10-pmsimstats-ng/pmsimstats-ng/analysis/scripts/informative-dropout-by-desi`